

Paper II - Urology (MCh Urology, Summer 2015)

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II-1. Describe medical management of BPH and role of various surgical modalities in its management.

Type: LAQ (25 marks)

Introduction / Definition

- **BPH (benign prostatic hyperplasia)** is a histological diagnosis: stromal and epithelial hyperplasia of the transition zone.
- Enlargement can cause **BPO (benign prostatic obstruction)**, presenting as **LUTS (lower urinary tract symptoms)**: storage, voiding and post-micturition.
- Goals: relieve bother and prevent progression to AUR (acute urinary retention) or surgery.

Pathophysiology relevant to therapy

- **Static component** - prostate bulk, dependent on DHT (dihydrotestosterone); targeted by 5-ARIs (5 α -reductase inhibitors) and surgery.
- **Dynamic component** - α 1A-adrenergic smooth-muscle tone; targeted by α -blockers.
- **Bladder response** - detrusor overactivity; targeted by antimuscarinics and β 3-agonists.

Evaluation

- **IPSS (International Prostate Symptom Score)**: 0-35; mild 0-7, moderate 8-19, severe 20-35.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)** and a 3-day FVC (frequency-volume chart).
- Qmax (maximum flow rate) <10 mL/s suggests BOO (bladder outlet obstruction); measure PVR (post-void residual).
- Before surgery, AUA (American Urological Association) advises measuring prostate size, because it decides the modality.

Medical management

1. **Watchful waiting** - for mild symptoms: fluid timing, less caffeine and alcohol.
2. **α 1-blockers** - first line for moderate-severe bothersome LUTS; work within days.

Drug	Dose	Note
Tamsulosin	0.4 mg once daily	α 1A/ α 1D selective
Alfuzosin	10 mg once daily	Least ejaculatory dysfunction
Silodosin	8 mg once daily	Most α 1A selective; highest anejaculation

- They improve IPSS by about 30–40% but **do not shrink the prostate or reduce AUR/surgery risk**.
- **IFIS (intraoperative floppy iris syndrome)** is strongest with tamsulosin; men planning cataract surgery should defer starting an α -blocker.

3. 5-ARIs – finasteride 5 mg or dutasteride 0.5 mg.

- Used when the prostate is enlarged (AUA: PSA >1.5 ng/mL or volume >30 mL; EAU (European Association of Urology): >40 mL).
- They shrink the prostate by about 20–25% over 6–12 months, halve PSA and reduce the risk of AUR and surgery by about 50%.
- Side effects: reduced libido, ED (erectile dysfunction), gynaecomastia.

4. Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: doxazosin plus finasteride reduced clinical progression by **66%** versus placebo (doxazosin 39%, finasteride 34%). Only finasteride or combination reduced AUR and invasive therapy.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination reduced AUR and surgery by about 66% versus tamsulosin.

5. Storage symptoms – add an antimuscarinic (caution if PVR >250–300 mL because of retention risk; cognitive side effects in the elderly) or a β 3-agonist (mirabegron 25–50 mg).

6. Tadalafil 5 mg daily – improves LUTS with or without ED; contraindicated with nitrates.

7. Nocturnal polyuria – low-dose desmopressin; monitor serum sodium.

LUTS → IPSS, DRE, urinalysis, PSA, FVC
 Mild → watchful waiting
 Voiding → α -blocker (+ 5-ARI if >30–40 mL)
 Storage persists → antimuscarinic or β 3-agonist
 Nocturnal polyuria → desmopressin
 Failure → surgery (choice by prostate size)

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency due to BPO, and LUTS failing medical therapy.

Modality	Prostate size (AUA)	Advantages	Disadvantages
TURP (transurethral resection of prostate)	30–80 mL	Reference standard	Bleeding, retrograde ejaculation (~65–75%), TUR syndrome, stricture
TUIP (transurethral incision of prostate)	≤30 mL	Less retrograde ejaculation	Higher retreatment
PVP (photoselective vaporization)	30–80 mL	Safe in anticoagulated men	No histology
HoLEP (holmium laser enucleation)	Size-independent	Lowest retreatment, least bleeding	Learning curve
Simple prostatectomy	>80–100 mL	Complete removal	Morbidity, transfusion
Aquablation	30–80; consider 80–150 mL	Preserves ejaculation	Bleeding
PUL (prostatic urethral lift)	30–80 mL, no median lobe	Preserves sexual function	Higher retreatment

- **TUR syndrome** with monopolar TURP: the hypo-osmolar irrigant (1.5% glycine) is absorbed, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with

resection >60–90 min. Treat by stopping surgery, diuretics and 3% saline if severe.

- **Bipolar TURP** uses saline and virtually eliminates TUR syndrome.

Landmark trials

- **GOLIATH**: PVP was non-inferior to TURP.
- **WATER (Waterjet Ablation Therapy for Endoscopic Resection of prostate tissue)**: Aquablation was non-inferior to TURP with less anejaculation.

Complications of surgery

- Early: bleeding, TUR syndrome, sepsis. Late: retrograde ejaculation, urethral stricture, bladder neck contracture.

Follow-up

- Reassess IPSS at 4–12 weeks on α -blockers and 3–6 months on 5-ARIs, then yearly.

Recent advances

- **Optilume BPH** (paclitaxel-coated balloon) was approved in 2023.
- Office-based MISTs (minimally invasive surgical therapies) offer ejaculation-preserving treatment.

Diagram to draw: Size-based algorithm: ≤ 30 mL TUIP $\rightarrow$ 30–80 mL TURP/PVP $\rightarrow$ > 80 mL enucleation or simple prostatectomy.

Key points to remember:

- IPSS 0–35: 0–7 mild, 8–19 moderate, 20–35 severe.
 - 5-ARIs shrink the prostate ~25%, halve PSA and cut AUR/surgery risk ~50%; MTOPS combination cut progression 66%.
 - AUA sizes: TUIP ≤ 30 ; TURP/PVP/PUL 30–80; HoLEP any size; simple prostatectomy > 80 –100 mL.
 - Bipolar TURP avoids TUR syndrome; GOLIATH: PVP non-inferior to TURP.
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II-2. Discuss metabolic evaluation in urinary calculus disease.

Type: LAQ (25 marks)

Introduction

- Urolithiasis is a systemic metabolic disease; about 50% of patients recur within 5–10 years without prevention.

Physicochemical basis

- Stones form when urine is **supersaturated**: nucleation, then growth, aggregation and retention.
- In the metastable zone, inhibitors prevent crystallisation; in the unstable zone, nucleation is spontaneous.
- Inhibitors include citrate and magnesium.
- **RSS (relative supersaturation)** is calculated from 24-h urine and used to guide therapy (target <1).

Who needs metabolic evaluation

- **Basic evaluation**: every stone former.
- **Comprehensive 24-h urine evaluation**: high-risk, recurrent and interested first-time stone formers.
- **High-risk features (EAU (European Association of Urology))**:
 - Children, family history, recurrent stones, solitary kidney.
 - Uric acid, infection or cystine stones.
 - Primary hyperparathyroidism, GI (gastrointestinal) disease, RTA (renal tubular acidosis) type 1, cystinuria.

Basic metabolic evaluation (all patients)

- **History**: diet (fluid, salt, animal protein), drugs and bowel disease.
- **Imaging**: NCCT (non-contrast computed tomography); uric acid stones are usually <500 HU (Hounsfield units).
- **Serum**: creatinine, calcium, uric acid, bicarbonate (low in RTA or diarrhoea) and phosphate. Check PTH (parathyroid hormone) if calcium is high-normal or raised.
- **Urine**: pH >6.5–7 suggests RTA or infection; pH <5.5 favours uric acid. Hexagonal crystals mean cystine.
- **Stone analysis** at the first episode by infrared spectroscopy or X-ray diffraction.

Comprehensive evaluation - 24-h urine

- Two collections on the usual diet, at least 2–4 weeks after the stone event, when stone-free and infection-free.
- Check completeness with urinary creatinine (men ~20–25, women ~15–20 mg/kg/day).

Parameter	Desirable (per day)	Abnormality
Volume	> 2.5 L	<2 L is the commonest finding
Calcium	<250 mg (women), <300 mg (men)	Hypercalciuria
Oxalate	< 40–45 mg	Hyperoxaluria
Citrate	> 320 mg	Hypocitraturia
Uric acid	<750 mg (women), <800 mg (men)	Hyperuricosuria

Parameter	Desirable (per day)	Abnormality
Sodium	<100 mmol	High salt raises calcium excretion
pH	5.8-6.2	<5.5 uric acid; >6.5-7 calcium phosphate, RTA, infection
Cystine	Very low	Cystinuria (>400 mg/day)

Interpretation and diagnostic categories

1. **Hypercalciuria** – absorptive, renal leak (raised PTH, normal serum calcium) or resorptive (primary hyperparathyroidism, with raised calcium and PTH). Thiazides work for all idiopathic types.
2. **Hyperoxaluria** – dietary (mild, 45–60 mg/day); enteric (in fat malabsorption, unbound fatty acids bind calcium, so free oxalate is absorbed in the colon; also hypocitraturia and low volume); primary hyperoxaluria (often >75–100 mg/day).
3. **Hypocitraturia** – idiopathic, distal RTA, chronic diarrhoea, hypokalaemia.
4. **Hyperuricosuria** promotes calcium oxalate stones by heterogeneous nucleation. **Uric acid stones** are driven mainly by urine pH <5.5 (metabolic syndrome, diabetes).
5. **Distal RTA** – urine pH stays >5.5–6, with hypocitraturia and calcium phosphate stones. In incomplete forms, an ammonium chloride load (0.1 g/kg) fails to lower urine pH below 5.5.
6. **Cystinuria** – autosomal recessive defect of cystine, ornithine, lysine and arginine (COLA); cystine solubility is ~250 mg/L at physiological pH, so aim for <250 mg/L.
7. **Infection stones** – urease-producing organisms, urine pH >7, struvite.

Stone episode → basic evaluation

Low risk → general advice (urine >2.5 L, less salt and protein, normal calcium)

High risk / recurrent → two 24-h urines → specific abnormality

→ targeted therapy → repeat 24-h urine at ~6 months → yearly

Treatment linked to metabolic abnormality

Abnormality	Diet	Drug
Hypercalciuria	Sodium <2.3 g/day, normal calcium, less animal protein	Thiazide (HCTZ (hydrochlorothiazide) 25–50 mg, chlorthalidone 25 mg, indapamide 2.5 mg) plus potassium citrate to prevent hypokalaemia
Primary hyperparathyroidism	–	Parathyroidectomy
Hypocitraturia, distal RTA	More fruit and vegetables	Potassium citrate 30–60 mEq/day
Enteric hyperoxaluria	Low fat and oxalate; calcium with meals	Calcium citrate, cholestyramine
Primary hyperoxaluria type 1	High fluid	Pyridoxine, lumasiran ; liver ± kidney transplant
Hyperuricosuria	Low purine	Allopurinol 100–300 mg
Uric acid stones	–	Potassium citrate to pH 6.0–6.5
Cystinuria	Fluid >3–4 L	Alkalise to pH 7.0–7.5 ; tiopronin if refractory
Infection stones	–	Complete stone clearance, culture-specific antibiotics

Special groups

- **Children:** all need full metabolic evaluation.
- **Pregnancy:** defer 24-h urine until after delivery and lactation, because pregnancy causes physiological hypercalciuria.

Follow-up

- Repeat 24-h urine within 6 months of starting therapy, then annually.

Recent advances

- **RNAi (ribonucleic acid interference) drugs** (lumasiran, nedosiran) markedly lower urinary oxalate in primary hyperoxaluria type 1.
- **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence versus placebo.

Diagram to draw: Supersaturation zones and the basic vs comprehensive evaluation algorithm.

Key points to remember:

- Two 24-h urines on usual diet, ≥ 4 weeks after event; check creatinine for adequacy.
 - Targets: volume >2.5 L, Ca $<250/300$ mg, oxalate <40 mg, citrate >320 mg, uric acid $<750/800$ mg.
 - Distal RTA: NH_4Cl load fails to lower pH <5.5 ; cystinuria: cystine <250 mg/L, pH 7.0–7.5, tiopronin.
 - Repeat 24-h urine within 6 months; lumasiran for primary hyperoxaluria type 1.
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II-3a. Management of priapism

Type: SAQ (10 marks)

Definition

- A full or partial erection lasting **>4 hours** beyond, or unrelated to, sexual stimulation (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2021).

Classification

Feature	Ischaemic (low-flow)	Non-ischaemic (high-flow)	Stuttering
Mechanism	Veno-occlusive compartment syndrome	Cavernous artery-sinusoidal fistula	Recurrent self-limiting ischaemic episodes
Cause	ICI (intracavernosal injection) drugs, SCD (sickle cell disease), antipsychotics	Perineal trauma	SCD
Penis	Rigid, painful	Partially rigid, painless	Painful
Emergency	Yes	No	Prevent progression

Diagnosis

- Corporal blood gas** in ischaemic priapism: pO₂ <30 mmHg, pCO₂ >60 mmHg, pH <7.25.
- Colour Doppler:** absent flow in ischaemic; high flow or fistula in non-ischaemic.

Management of ischaemic priapism

Dorsal penile block → aspiration (16–18G) ± saline irrigation
→ Phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to ~1 h
→ Failure → distal shunt → proximal shunt
→ >36 h or failed shunts → immediate penile prosthesis

- Distal (cavernoglanular) shunts** drain the corpora into the glans: Winter, Ebbehøj, T-shunt, Al-Ghorab.
- Proximal shunts:** Quackels (cavernoso-spongiosal), Grayhack (cavernoso-saphenous).
- Ischaemia >24–48 h causes fibrosis and ED (erectile dysfunction); early prosthesis is easier and preserves length, whereas delayed implantation into fibrotic corpora is difficult.
- In SCD, give intracavernosal therapy alongside haematology care.

Non-ischaemic priapism

- Not an emergency; observe first.
- If persistent, use selective embolisation; absorbable gelfoam or autologous clot carries lower ED risk than permanent coils.

Stuttering priapism

- Treat each episode as ischaemic.
- Prevent with daily PDE5i (phosphodiesterase-5 inhibitor, which acts paradoxically when taken in the non-erect state), pseudoephedrine or self-injected phenylephrine; hydroxyurea in SCD.

Recent advances

- Early penile prosthesis in refractory cases.

Diagram to draw: Penile cross-section showing distal vs proximal shunts.

Key points to remember:

- Ischaemic: $pO_2 < 30$, $pCO_2 > 60$, $pH < 7.25$ - emergency.
 - Phenylephrine 100-500 $\mu\text{g/mL}$, 1 mL q3-5 min up to 1 h.
 - Distal shunts (Winter, Ebbehøj, T, Al-Ghorab) before proximal (Quackels, Grayhack).
 - Prosthesis for > 36 h; high-flow $\rightarrow$ observe/embolise.
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II-3b. Renal tubular acidosis

Type: SAQ (10 marks)

Definition

- **RTA (renal tubular acidosis)** is a hyperchloraemic, normal AG (anion gap) metabolic acidosis caused by impaired bicarbonate reabsorption or acid excretion,.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	Impaired H ⁺ secretion in collecting duct	Impaired HCO ₃ reabsorption in proximal tubule	Aldosterone deficiency or resistance
Urine pH	Always >5.5	<5.5 once serum HCO ₃ falls below the lowered threshold; >5.5 during bicarbonate loading	<5.5
Serum potassium	Low	Low	High
Urine citrate	Low	Normal/high	Normal
Stones/nephrocalcinosis	Common (calcium phosphate)	Rare	Absent
Treatment	Potassium citrate 1-2 mEq/kg/day	Alkali 10-15 mEq/kg/day plus potassium	Fludrocortisone , low-potassium diet

- **Incomplete distal RTA:** normal serum bicarbonate but the urine cannot be acidified; common in recurrent calcium phosphate stone formers.

Diagnosis

- **UAG (urine anion gap) = (Na + K) - Cl** is a surrogate for urinary ammonium excretion. Negative UAG means GI (gastrointestinal) loss or proximal RTA; positive UAG means distal or type 4 RTA.
- **Ammonium chloride test** (0.1 g/kg): failure to lower urine pH below 5.5 confirms distal RTA.
- Fractional excretion of bicarbonate >15% during bicarbonate loading indicates proximal RTA.

Urological relevance

- Distal RTA causes recurrent calcium phosphate stones, nephrocalcinosis and hypocitraturia.
- Obstructive uropathy (posterior urethral valves, prostatic obstruction) causes type 4 RTA, which improves after relief.
- Urinary diversion causes hyperchloraemic acidosis that mimics RTA.
- Potassium citrate is preferred over sodium bicarbonate, because sodium increases calcium excretion.

Recent advances

- Genetic testing for inherited distal RTA; prolonged-release alkali formulations.

Diagram to draw: Nephron showing the defect sites: proximal tubule (type 2), α -intercalated cell (type 1), aldosterone action (type 4).

Key points to remember:

- Type 1: urine pH >5.5, hypokalaemia, hypocitraturia, CaP stones/nephrocalcinosis.
 - Type 4: hyperkalaemia, no stones; seen in obstructive uropathy.
 - Positive UAG = distal RTA; NH₄Cl 0.1 g/kg – pH fails to fall <5.5.
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II-3c. Millin's prostatectomy

Type: SAQ (10 marks)

Definition / History

- A **retropubic extravesical simple prostatectomy**: the adenoma is enucleated through a transverse incision in the anterior prostatic capsule, without opening the bladder.
- Described by Terence Millin in 1945. In contrast, **Freyer's** transvesical prostatectomy removes the adenoma through an opened bladder.

Indications

- Large glands, generally **>80-100 mL** (AUA (American Urological Association)), when HoLEP (holmium laser enucleation of prostate) is unavailable.
- Relative contraindications: large median lobe, diverticulum or large stones (transvesical preferred), prostate cancer.

Steps

1. Pfannenstiel incision; extraperitoneal entry into the **space of Retzius**.
2. Haemostatic sutures across the anterior capsule to control the Santorini plexus.
3. **Transverse capsulotomy** about 1-2 cm distal to the bladder neck.
4. **Finger enucleation** of the adenoma from the surgical capsule.
5. **Divide the urethra at the apex under vision** to protect the sphincter.
6. **Wedge-excise the posterior bladder neck** to prevent contracture.
7. Haemostatic sutures at **5 and 7 o'clock**.
8. 20-22 Fr three-way catheter, **watertight capsule closure** and drain.

Advantages over Freyer (transvesical)

Aspect	Millin	Freyer
Bladder	Not opened	Opened
Apex	Divided under vision – better continence	Blind finger fracture
Stones, diverticulum, large median lobe	Difficult	Ideal
Specific risk	Santorini bleeding, osteitis pubis	Bladder spasm, suprapubic fistula

Complications

- Haemorrhage needing transfusion, infection, retrograde ejaculation (most patients), bladder neck contracture, osteitis pubis.

Modern alternatives

- **HoLEP** is size-independent with less bleeding; EAU (European Association of Urology) recommends it first for >80 mL where available.
- **RASP (robotic-assisted simple prostatectomy)** gives less blood loss than open surgery.

Diagram to draw: Sagittal section showing the capsulotomy below the bladder neck and the enucleation plane.

Key points to remember:

- Millin 1945: retropubic, transverse anterior capsulotomy, bladder not opened.
 - Apical urethral division under vision; 5 and 7 o'clock sutures; bladder neck wedge.
 - Freyer (transvesical) preferred for median lobe, stones, diverticulum.
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II-3d. Energy sources in laparoscopic urology

Type: SAQ (10 marks)

Introduction

- Energy devices cut, coagulate and seal vessels.
- Tissue effects: about 60 °C denatures protein (coagulation), about 100 °C vaporizes water (cutting), above 200 °C carbonizes.

Types of energy sources

Device	Principle	Use	Limitation
Monopolar	RF (radiofrequency) current passes through the patient to a return pad; continuous low-voltage waveform cuts, interrupted high-voltage waveform coagulates	Cheap, versatile	Greatest lateral spread; stray current
Bipolar	Current flows only between the jaws	Safer near nerves	Slow; no cutting
Advanced bipolar (LigaSure)	Pressure plus impedance-controlled energy fuses collagen and elastin	Seals vessels up to 7 mm	Cost; spread ~1-3 mm
Ultrasonic (Harmonic)	Blade vibrates at ~55.5 kHz; mechanical energy denatures protein below 100 °C	Cuts and seals vessels up to 5 mm ; no current through patient	Blade stays hot
Argon beam	Monopolar current through argon gas, non-contact	Haemostasis of the partial nephrectomy bed	Gas embolism

Safety - mechanisms of electrosurgical injury

- Injuries outside the field of view present later as bowel perforation. 1. **Insulation failure** - current leaks through damaged insulation. 2. **Direct coupling** - the active electrode touches another metal instrument. 3. **Capacitive coupling** - current is induced through intact insulation, worse with high-voltage coagulation; avoid hybrid (metal-plastic) trocars. 4. **Thermal spread** - to ureter, nerves or bowel. 5. **Return electrode burns** - from poor pad contact.

Safety measures

- **AEM (active electrode monitoring)** shuts the generator if insulation fails or capacitive coupling occurs.
- **REM (return electrode contact quality monitoring)** prevents pad burns.
- Use the lowest effective power, keep the tip in view, and prefer bipolar near critical structures.

Recent advances

- Smart generators with real-time impedance feedback; robotic vessel sealers.

Diagram to draw: Monopolar vs bipolar circuit; capacitive coupling.

Key points to remember:

- LigaSure: vessels up to 7 mm; Harmonic: up to 5 mm at ~55.5 kHz.
- Stray current injury: insulation failure, direct coupling, capacitive coupling.
- Avoid hybrid trocars; use AEM and REM pads.

II-3e. Role of vascular grafts in urology

Type: SAQ (10 marks)

Introduction

- Vascular grafts replace, bypass or extend vessels when primary repair is impossible.

Types of grafts

Type	Examples	Advantages	Disadvantages
Autologous vein	Saphenous, gonadal	Low infection, good small-vessel patency	Limited length
Autologous artery	Internal iliac, inferior epigastric, Carrel aortic patch	Ideal for renal artery reconstruction	Limited availability
Synthetic	PTFE (polytetrafluoroethylene) , Dacron	Off-the-shelf; for IVC (inferior vena cava) and aorta	Infection; avoid in contaminated fields

Applications in urology

1. **Renal transplantation:** multiple donor arteries are joined on the bench (side-to-side or end-to-side); a short right renal vein is extended with gonadal or IVC segments.
2. **Renovascular surgery:** renal artery stenosis or aneurysm is treated by aortorenal bypass using saphenous vein, hypogastric artery or PTFE.
3. **RCC (renal cell carcinoma) with IVC thrombus:** most need cavotomy and primary closure. Patch cavoplasty is used if the lumen is narrowed >50%; PTFE IVC replacement if the wall is invaded and collaterals are inadequate.
4. **Post-chemotherapy RPLND (retroperitoneal lymph node dissection):** an encased aorta may need a Dacron or PTFE interposition graft.
5. **Dialysis access:** PTFE AV (arteriovenous) graft.

Principles

- Prefer autologous grafts in contaminated or urine-exposed fields and small vessels; synthetic grafts for large high-flow vessels.
- Make tension-free, spatulated anastomoses; heparinise during clamping.
- Wrap synthetic grafts in omentum to separate them from bowel and urinary tract.
- Infrarenal IVC can often be ligated without a graft when collaterals have developed.

Complications

- Thrombosis, stenosis, pseudoaneurysm, graft infection (often needing explantation), graft-enteric or urinary fistula.

Recent advances

- Robotic IVC thrombectomy; human acellular vessels for dialysis access and trauma.

Diagram to draw: Bench reconstruction of two renal arteries before transplant.

Key points to remember:

- Autologous (saphenous, hypogastric, gonadal) for small vessels/contaminated fields; PTFE/Dacron for IVC/aorta.
 - Transplant: Carrel patch, bench reconstruction, gonadal/IVC extension of short right vein.
 - IVC: patch cavoplasty if narrowed; PTFE replacement if wall invaded.
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II-3f. Transrectal ultrasonography

Type: SAQ (10 marks)

Introduction / Physics

- Piezoelectric crystals convert electrical energy to sound and back; higher frequency gives better resolution but less penetration.
- Probes are **6-10 MHz (commonly 7.5 MHz)** and biplanar (transverse and sagittal).

Technique

- Left lateral position; DRE (digital rectal examination) first, then transverse and sagittal scanning.

Normal anatomy (McNeal zones)

Zone	Glandular volume	Echogenicity	Share of cancers
Peripheral	~70%	Isoechoic	~70%
Transition	~5-10% (enlarges in BPH (benign prostatic hyperplasia))	Hypoechoic	~20-25%
Central	~25%	Slightly hyperechoic	~5-10%

Measurements

- **Prostate volume** = $0.52 \times \text{length} \times \text{width} \times \text{height}$.
- **PSAD (PSA (prostate-specific antigen) density)** = PSA/volume; >0.15 is suspicious and helps decide biopsy after MRI (magnetic resonance imaging).
- Classic cancer is a hypoechoic peripheral-zone lesion, but specificity is low.

Indications

- Guiding prostate biopsy and measuring volume before BPH surgery.
- Ejaculatory duct obstruction: seminal vesicles >1.5 cm, midline cysts.
- Guidance for abscess drainage and brachytherapy.

TRUS (transrectal ultrasound)-guided biopsy

- Systematic **12-core** biopsy under a periprostatic nerve block.
- Complications: haemospermia (most common), haematuria and **sepsis** from FQ (fluoroquinolone)-resistant E. coli.
- Prophylaxis: povidone-iodine rectal cleansing and targeted prophylaxis based on rectal swab culture; fluoroquinolones are now restricted.

Current shift

- **EAU (European Association of Urology) 2024/25** recommends transperineal biopsy because of its lower infection risk.
- **Pre-biopsy mpMRI (multiparametric MRI)** with targeted biopsy:
 - **PROMIS (2017)**: mpMRI sensitivity for csPCa (clinically significant prostate cancer) was 93% versus 48% for TRUS biopsy.
 - **PRECISION (2018)**: MRI-targeted biopsy detected more csPCa (38% vs 26%) and less insignificant cancer.

Recent advances

- **Micro-ultrasound (29 MHz):** the OPTIMUM RCT (randomised controlled trial) found it non-inferior to MRI-targeted biopsy.

Diagram to draw: Transverse and sagittal views of zonal anatomy; 12-core biopsy template.

Key points to remember:

- Biplanar probe 6–10 MHz; volume = $0.52 \times L \times W \times H$; PSAD >0.15 suspicious.
 - 12-core systematic biopsy with periprostatic block.
 - EAU: transperineal preferred (lower sepsis); FQ prophylaxis restricted.
 - PROMIS and PRECISION established MRI-first pathway.
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Rapid Revision Sheet - Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

II-1. Describe medical management of BPH and role of various surgical modalities in its management. (LAQ)

- IPSS 0–35: 0–7 mild, 8–19 moderate, 20–35 severe.
- 5-ARIs shrink the prostate ~25%, halve PSA and cut AUR/surgery risk ~50%; MTOPS combination cut progression 66%.
- AUA sizes: TUIP ≤ 30 ; TURP/PVP/PUL 30–80; HoLEP any size; simple prostatectomy >80–100 mL.
- Bipolar TURP avoids TUR syndrome; GOLIATH: PVP non-inferior to TURP.

II-2. Discuss metabolic evaluation in urinary calculus disease. (LAQ)

- Two 24-h urines on usual diet, ≥ 4 weeks after event; check creatinine for adequacy.
- Targets: volume >2.5 L, Ca <250/300 mg, oxalate <40 mg, citrate >320 mg, uric acid <750/800 mg.
- Distal RTA: NH_4Cl load fails to lower pH <5.5; cystinuria: cystine <250 mg/L, pH 7.0–7.5, tiopronin.
- Repeat 24-h urine within 6 months; lumasiran for primary hyperoxaluria type 1.

II-3a. Management of priapism (SAQ)

- Ischaemic: $\text{pO}_2 < 30$, $\text{pCO}_2 > 60$, pH <7.25 – emergency.
- Phenylephrine 100–500 $\mu\text{g/mL}$, 1 mL q3–5 min up to 1 h.
- Distal shunts (Winter, Ebbehøj, T, Al-Ghorab) before proximal (Quackels, Grayhack).
- Prosthesis for >36 h; high-flow $\rightarrow$ observe/embolise.

II-3b. Renal tubular acidosis (SAQ)

- Type 1: urine pH >5.5, hypokalaemia, hypocitraturia, CaP stones/nephrocalcinosis.
- Type 4: hyperkalaemia, no stones; seen in obstructive uropathy.
- Positive UAG = distal RTA; NH_4Cl 0.1 g/kg – pH fails to fall <5.5.

II-3c. Millin's prostatectomy (SAQ)

- Millin 1945: retropubic, transverse anterior capsulotomy, bladder not opened.
- Apical urethral division under vision; 5 and 7 o'clock sutures; bladder neck wedge.
- Freyer (transvesical) preferred for median lobe, stones, diverticulum.

II-3d. Energy sources in laparoscopic urology (SAQ)

- LigaSure: vessels up to 7 mm; Harmonic: up to 5 mm at ~55.5 kHz.
- Stray current injury: insulation failure, direct coupling, capacitive coupling.
- Avoid hybrid trocars; use AEM and REM pads.

II-3e. Role of vascular grafts in urology (SAQ)

- Autologous (saphenous, hypogastric, gonadal) for small vessels/contaminated fields; PTFE/Dacron for IVC/aorta.
- Transplant: Carrel patch, bench reconstruction, gonadal/IVC extension of short right vein.
- IVC: patch cavoplasty if narrowed; PTFE replacement if wall invaded.

II-3f. Transrectal ultrasonography (SAQ)

- Biplanar probe 6–10 MHz; volume = $0.52 \times L \times W \times H$; PSAD >0.15 suspicious.
- 12-core systematic biopsy with periprostatic block.
- EAU: transperineal preferred (lower sepsis); FQ prophylaxis restricted.

- PROMIS and PRECISION established MRI-first pathway.